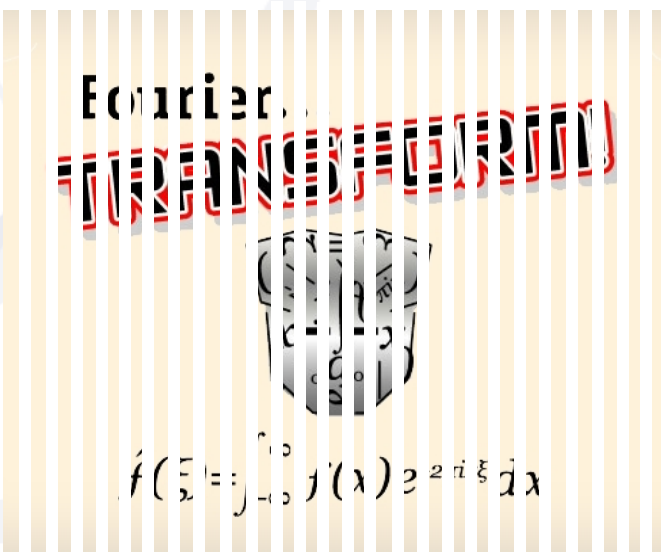
Signals and Systems

Module 10
Discrete Fourier Transform



Overview

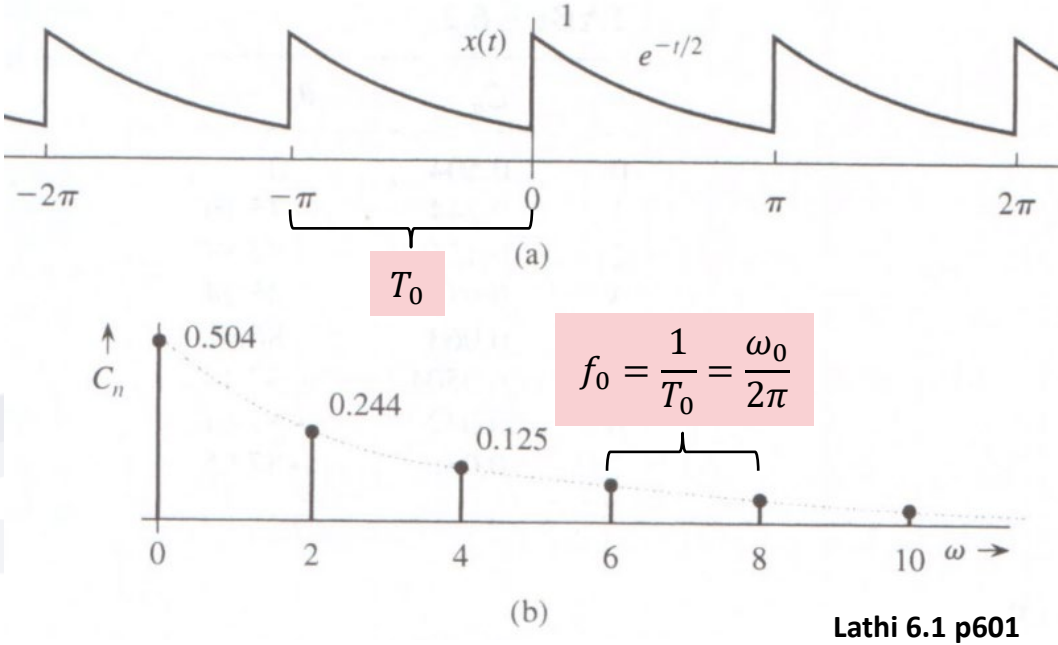
- This module will cover the following topics:
  - Spectral sampling
  - The Discrete Fourier transform (DFT)
  - Sources of error in DFT
  - Some applications of DFT
  - Fast Fourier Transform (FFT)

Recap: Fourier Series

- Any periodic signal (almost) can be represented as an infinite sum of sinusoids via the Fourier series:

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}$$

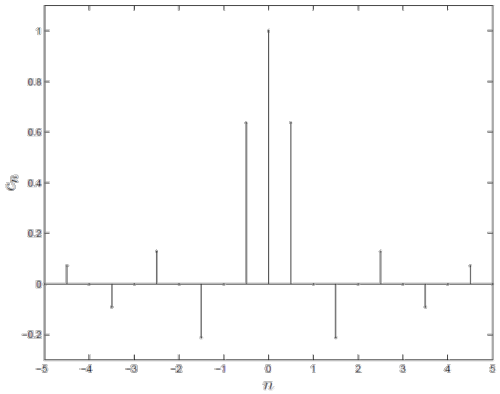
$$D_n = \frac{1}{T_0}\int_0^{T_0} x(t)e^{-jn\omega_0 t}dt$$



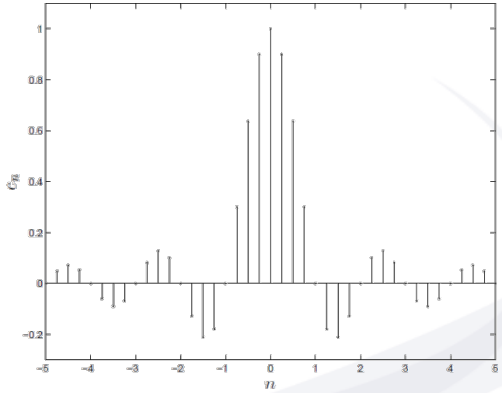
Lathi 6.1 p601

Recap: Aperiodic Signal Representation

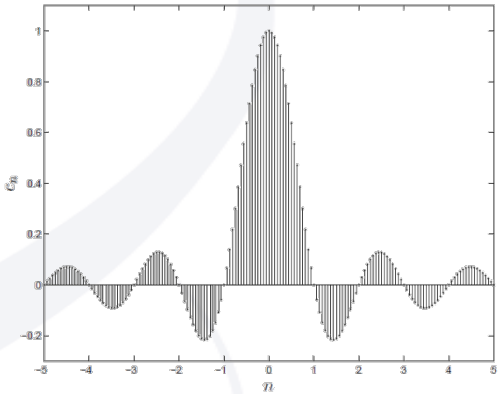
- Aperiodic signal – viewed as periodic signal with infinite period



Period = 2s



Period = 4s

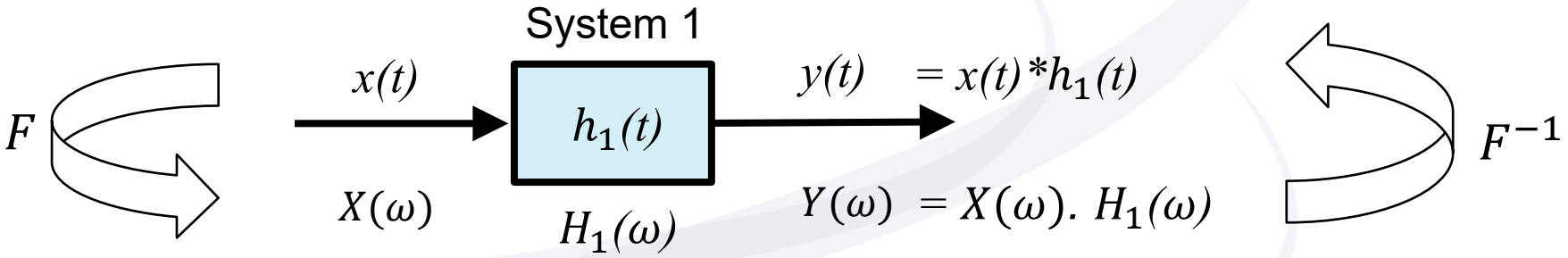


Period = 16s

- As $T_0 \to \infty, f_0 \to 0$
- End up with continuous spectrum (***Fourier Transform***)

Why use Fourier Transforms?

- To go from time domain to frequency domain for aperiodic signals
- And back again using inverse Fourier Transform



Recap: The Fourier Transform

Fourier Transform

Time domain → frequency domain

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

- Using ω (rad/s)

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt$$

- Using f (Hz)

Inverse Fourier Transform

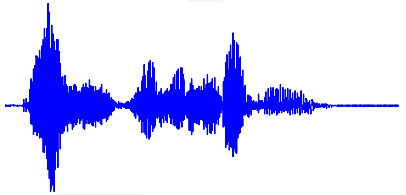
Frequency domain → time domain

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

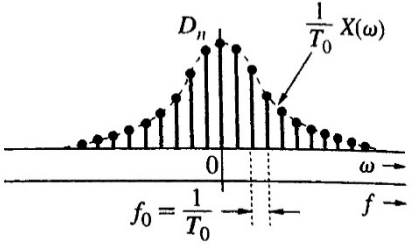
$$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi f t} df$$

Practical limitations of Fourier Transform

- Real-world signals don't always follow 'nice' mathematical functions
  - Not easy to integrate



- Numerical methods (using computers) require discrete values
- Even discrete time (sampled) signals could end up producing (repeated) continuous spectra

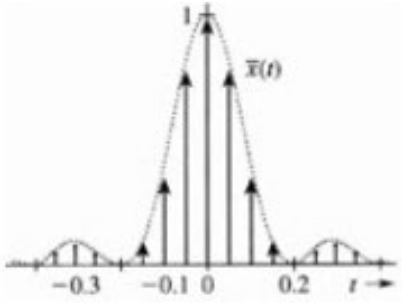


- Solution: **Spectral Sampling**

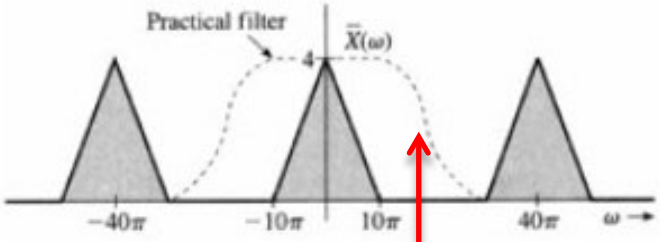
Recap: Time domain sampling

Sampled signal spectrum is replicated every f_s (sampling frequency)

Sampling higher than Nyquist rate

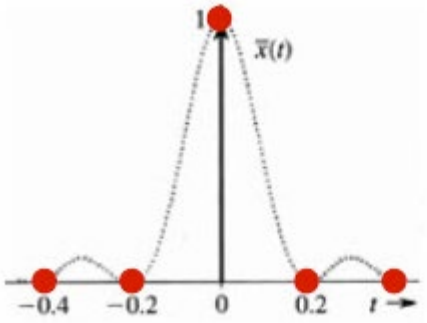


$\Leftrightarrow$



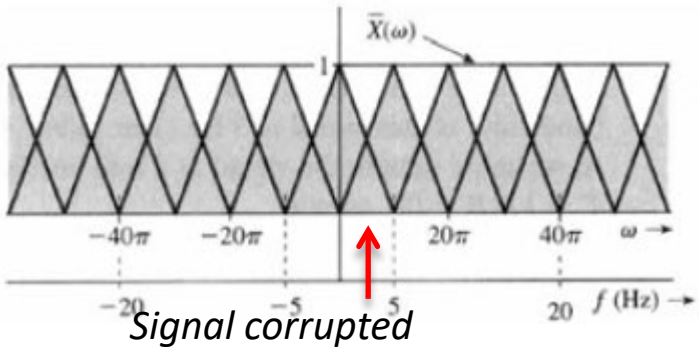
Perfect reconstruction possible with practical filter

Sampling below Nyquist rate



$\Leftrightarrow$

ALIASING

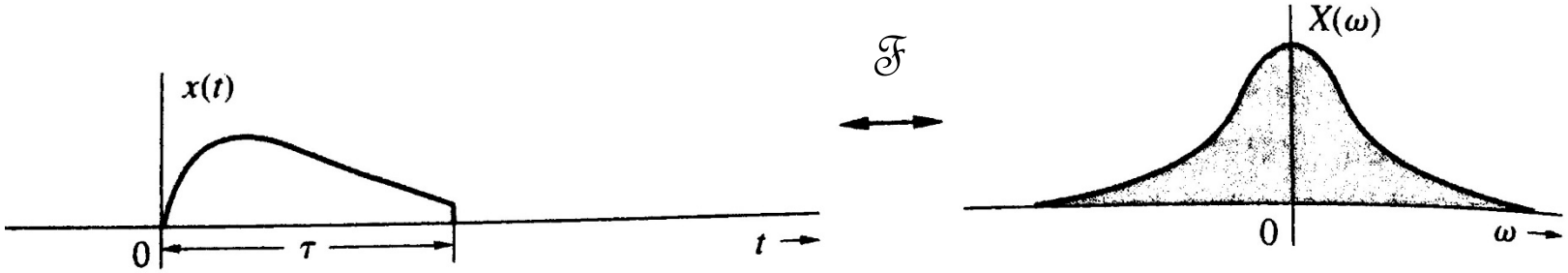


Signal corrupted

Lathi 8.1 p774

Spectral sampling

- The *dual* of time sampling
- Consider a time-limited signal *x(t)* with spectrum *X(ω)*

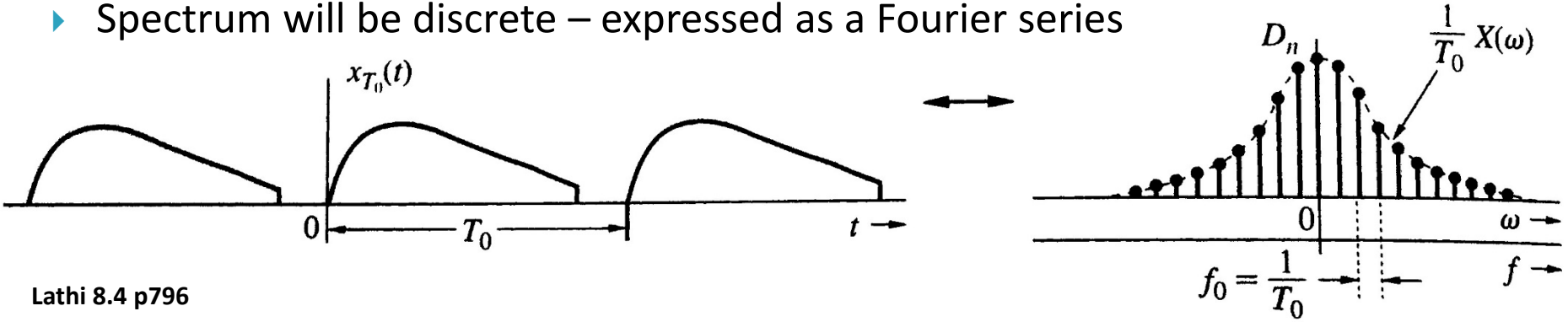


$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt = \int_{0}^{\tau} x(t)e^{-j\omega t}dt$$

Lathi 8.4 p796

Spectral sampling

- Construct a periodic signal $x_{T_0}(t)$: Repeat *x(t)* every T_0 seconds, $T_0 > \tau$
- Spectrum will be discrete – expressed as a Fourier series



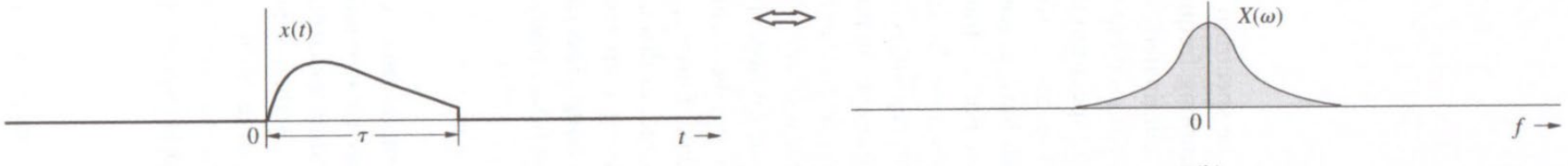
Lathi 8.4 p796

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t} \; ; \; \omega_0 = \frac{2\pi}{T_0} \qquad D_n = \frac{1}{T_0}\int_0^{T_0} x(t)e^{-jn\omega_0 t}dt = \frac{1}{T_0}\int_0^{\tau} x(t)e^{-jn\omega_0 t}dt$$

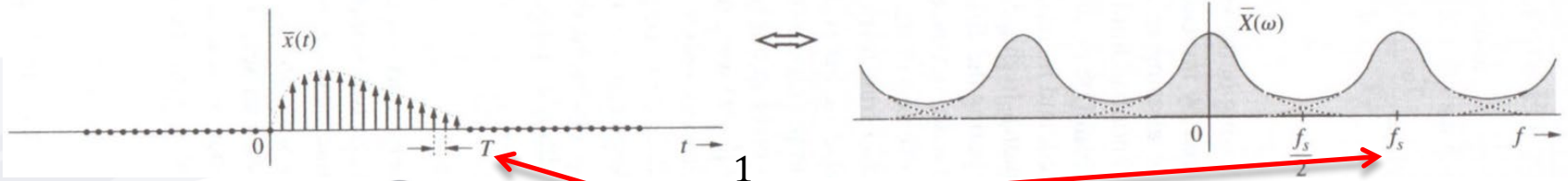
Therefore $$D_n = \frac{1}{T_0}X(n\omega_0)$$

The Discrete Fourier Transform (DFT)

- Numerical computation of Fourier transform (using computers)
- Given a time-limited signal *x(t)* want to compute its Fourier Transform *X(ω)*

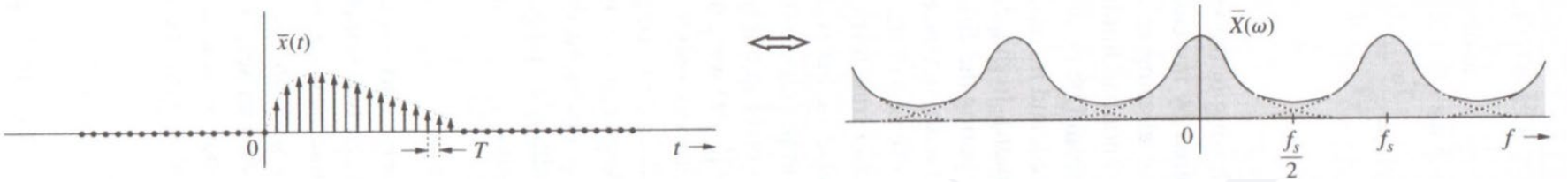


- Sampling the signal in time-domain

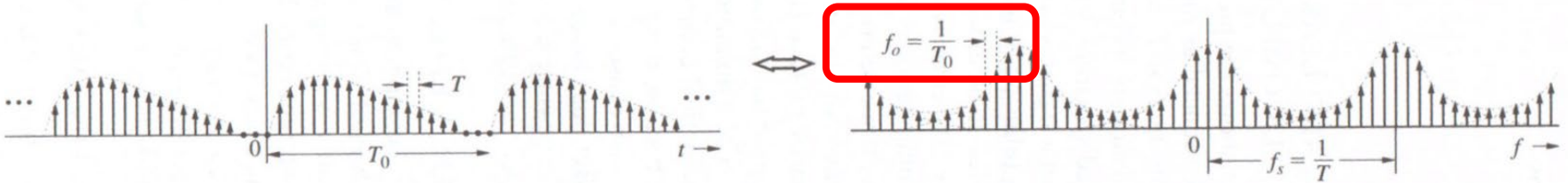


$$\frac{1}{T} = f_s$$

The Discrete Fourier Transform (DFT)



- Repeat sampled signal every T_0 seconds to create a periodic signal



Number of time samples in T_0

$$N_0 = \frac{T_0}{T}$$

Number of frequency samples in f_s

$$N_0' = \frac{f_s}{f_0}$$

$$N_0 = \frac{T_0}{T} = \frac{1/f_0}{1/f_s} = \frac{f_s}{f_0} = N_0'$$

Formal definition of DFT

- If $x(nT)$ is n^{th} sample of $x(t)$ and $X(r\omega_0)$ is r^{th} sample of $X(\omega)$, define

$$x_n = Tx(nT) = \frac{T_0}{N_0}x(nT) \quad \text{and} \quad X_r = X(r\omega_0) \quad \text{where} \quad \omega_0 = 2\pi f_0 = \frac{2\pi}{T_0}$$

- Then

DFT

$$X_r = \sum_{n=0}^{N_0-1} x_n e^{-jr\Omega_0 n}$$

Inverse DFT

$$x_n = \frac{1}{N_0}\sum_{r=0}^{N_0-1} X_r e^{jr\Omega_0 n}$$

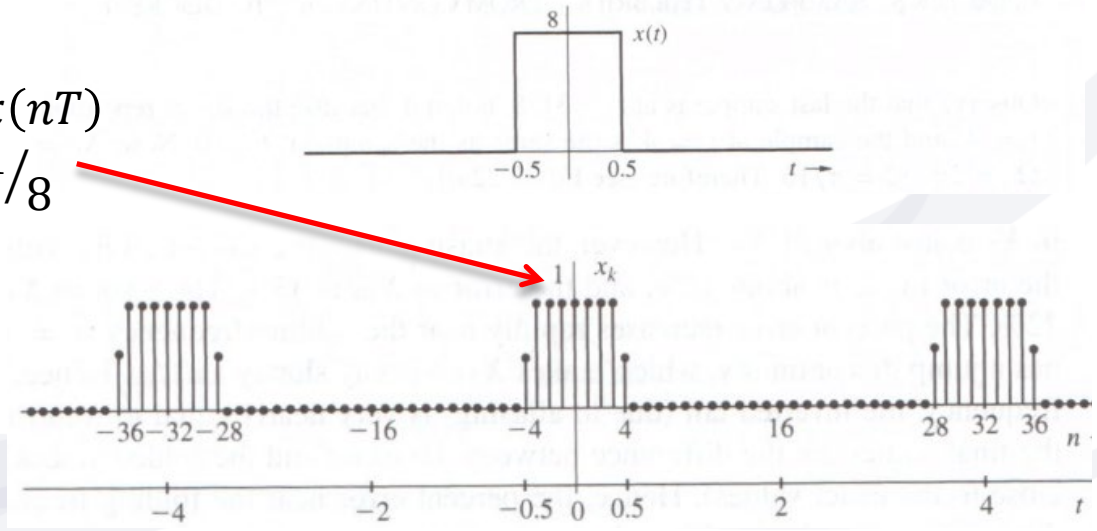
$$\Omega_0 = \omega_0 T = \frac{2\pi}{N_0}$$

Points of discontinuity

- If *x(t)* or *X(ω)* has a jump discontinuity at a sampling point, sample value should be average of values on either side

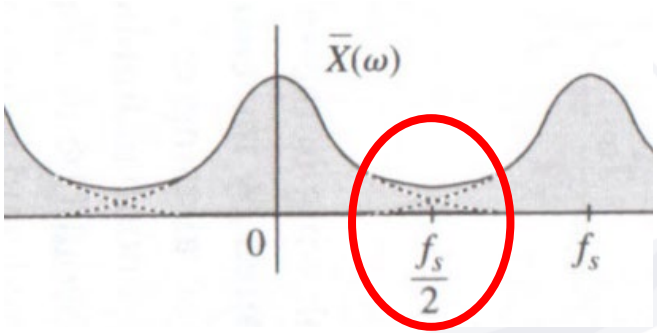
$x_n = Tx(nT)$

$T = {}^{1}/_{8}$



Aliasing in computation

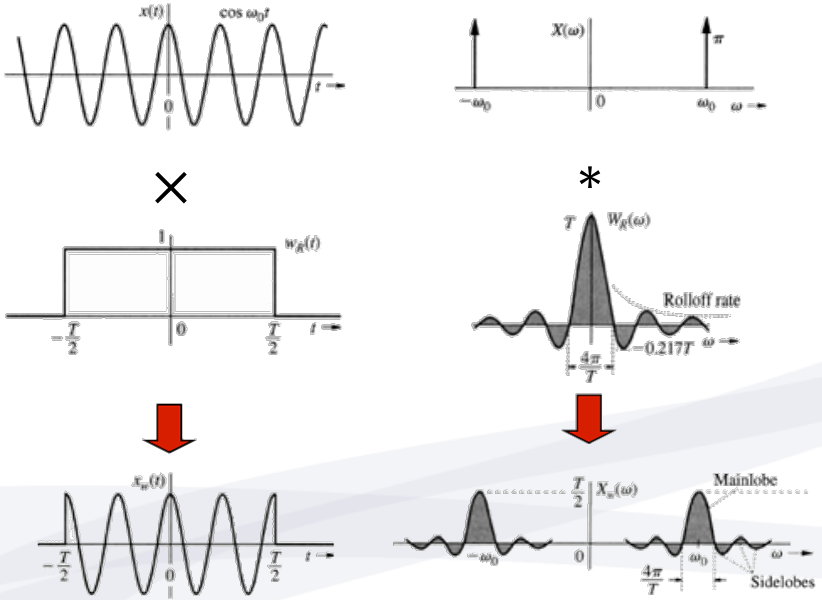
- Given that *x(t)* is time-limited, *X(ω)* is not bandlimited
- Since *X(ω)* is not bandlimited, get some aliasing effects



- Inherent error in computation of DFT
- Can be reduced by increasing the sampling frequency f_s
  - i.e. decreasing sampling interval T (= $1/f_s$)

Recap: Data Truncation – Windowing

- Signal truncation: multiply signal by a window function of smaller (finite) width



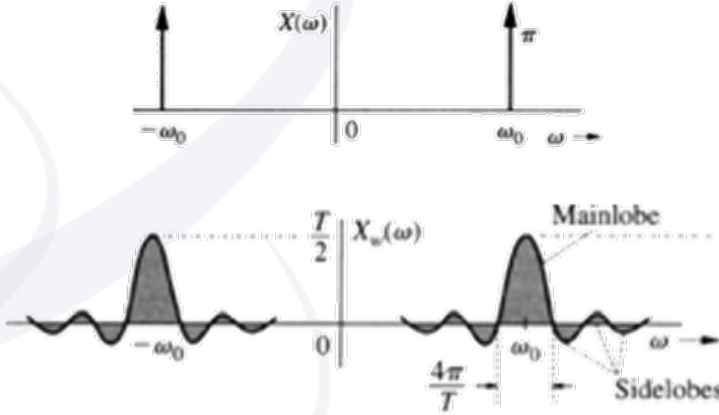
Problems:

Spectral Spreading: bandwidth increased by the bandwidth of $w(t)$ (loss of spectral resolution)

Leakage: $W(\omega)$ not bandlimited, therefore $X_w(\omega)$ spreads into the band where it should be zero

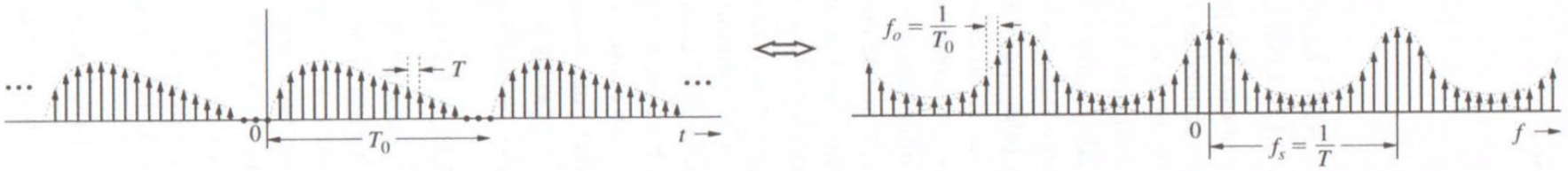
Leakage effects

- If *x(t)* is not time-limited, need to truncate *x(t)* with a window function
  - In practice, signal needs to ‘start’ and ‘finish’
- Leads to *leakage* and *spectral spreading*
- Leakage also causes aliasing
  - Can be reduced by using tapered window
  - But this increases spectral smearing!



Spectral spreading

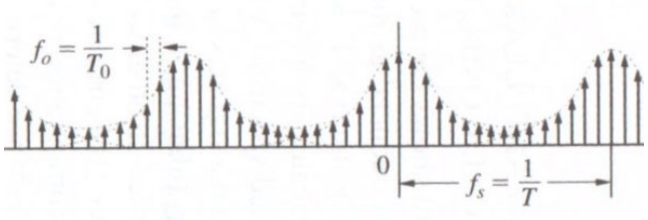
- Also known as spectral smearing
- Can be reduced by increasing the window width
- This increases T_0 and reduces f_0 (better frequency resolution)

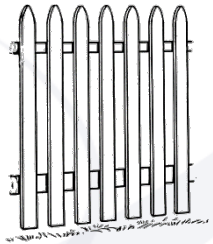


- *Zero padding*: adding dummy samples to increase T_0
  - Better frequency resolution
  - But does not really add any information!
  - Better view of distorted information

Picket fence effect

- Numerical computation yields uniform sampling values of $X(\omega)$
- Information between samples missing – picket fence effect

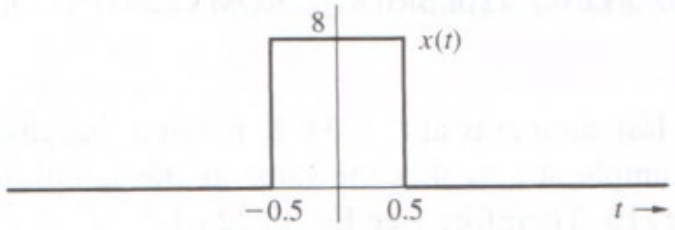


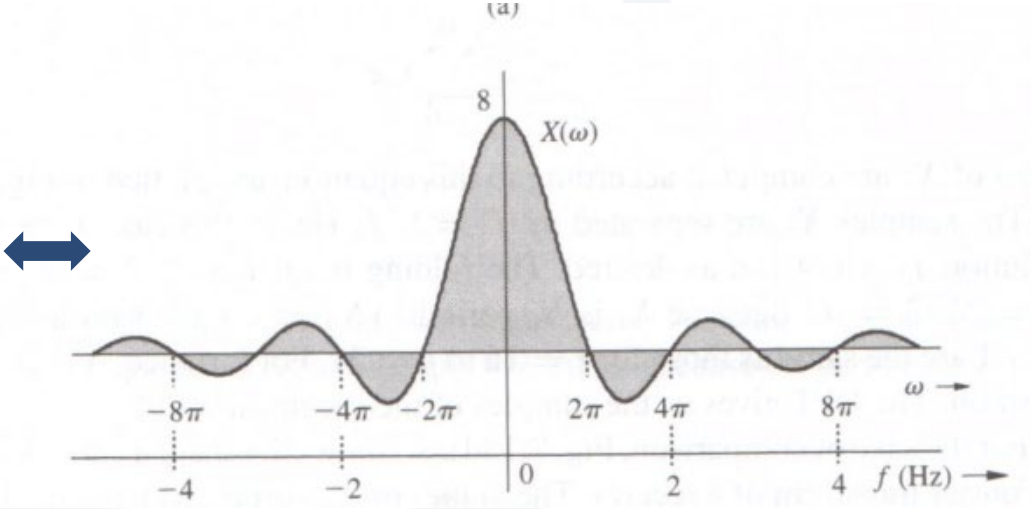


- Can use spectral interpolation methods to determine values in between
- Increase information by using sufficiently large number of samples N_0
  - Proper way is to reduce sampling interval T

Example

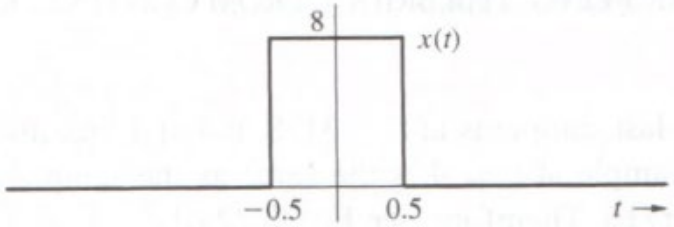
- Use DFT to compute Fourier transform of 8*rect(t)

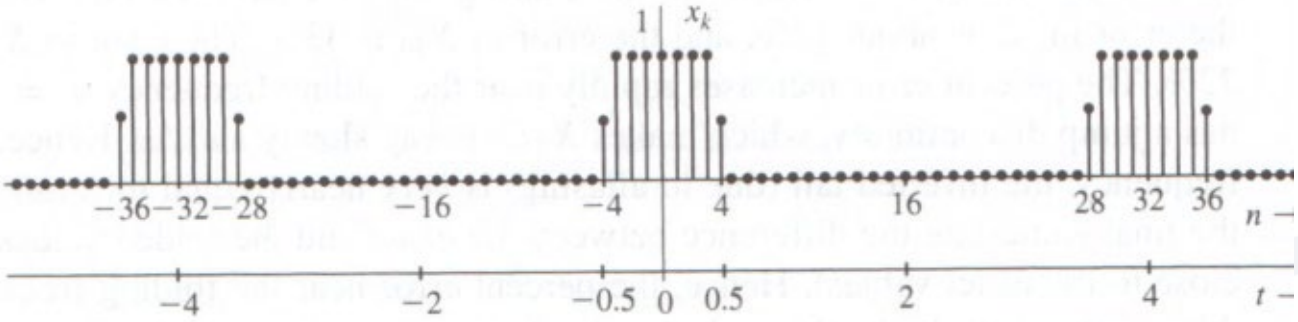




Example

- After sampling and repetition





Chosen f_s = 8 Hz → T = 1/8 s

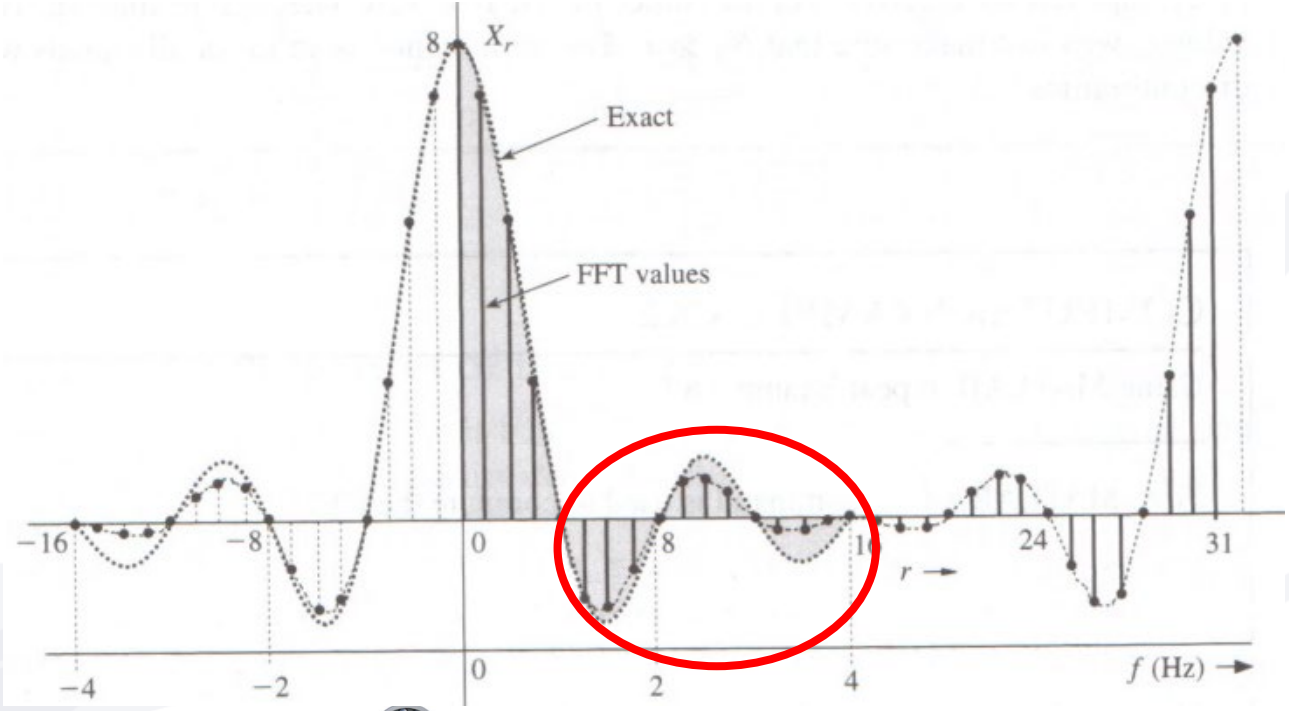
Want f_0 = 1/4 Hz → T_0 = 4 s

$\therefore N_0 = T_0 / T = 32$

Note: N_0 normally chosen as power of 2 (2^x) for computation reasons

Example

- Result of DFT



f_s = 8 Hz

f_0 = 1/4 Hz

$\therefore N_0$ = 32

Question 1

A signal is going to be evaluated using a 64-point DFT and the required resultant spectral resolution is 1/16 Hz. What is the required time domain sampling period?

1. ½ sec
2. ¼ sec
3. 1/8 sec
4. 1/16 sec
5. 1/64 sec

Web Address: http://pollev.com/signals

Question 1 – Solution

Given $f_0 = \frac{1}{16}$ and $N_0 = 64$

$$N_0 = \frac{f_s}{f_0} = 64 \qquad \therefore f_s = 64 \cdot \frac{1}{16} = 4$$

Sampling period, $T = \frac{1}{f_s} = \frac{1}{4} = 0.25$ s

Revisiting DT Systems & Convolution

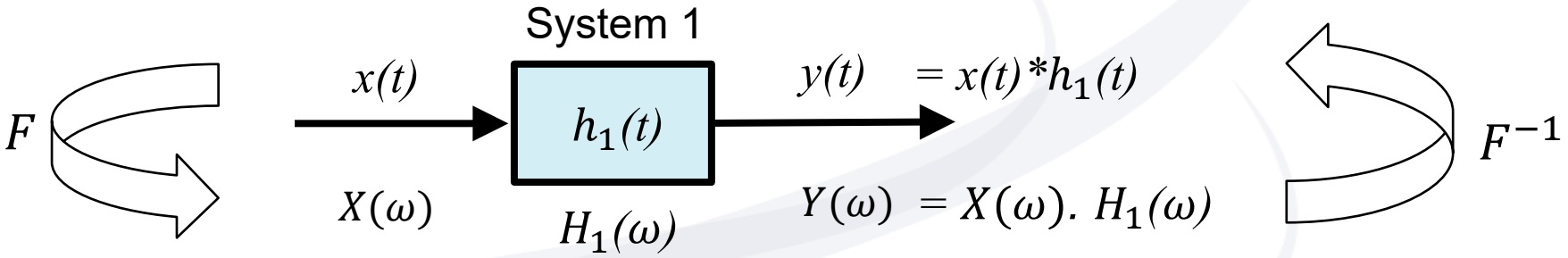
- DT system characteristic modes determined by solving the characteristic difference equation
- In discrete time, convolution is performed by the convolution sum:

$$y[n] = x[n] * h[n] = \sum_{m=-\infty}^{\infty} x[m]h[n-m]$$

- A DT LTI system is internally stable if and only if all of the characteristic roots are inside the unit circle

Why use Fourier Transforms?

- To go from time domain to frequency domain for aperiodic signals
- And back again using inverse Fourier Transform

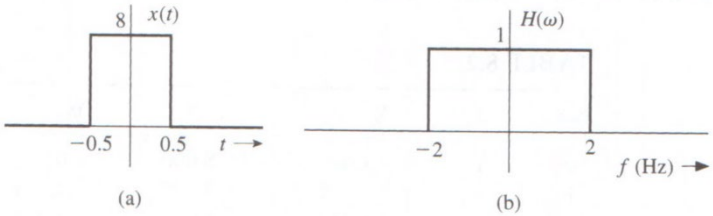


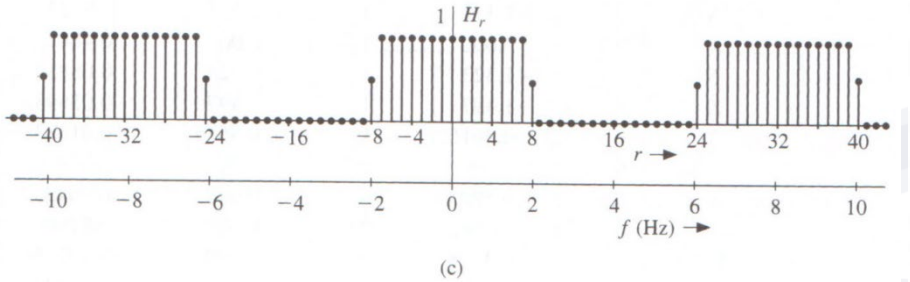
Applications of DFT – Linear Convolution

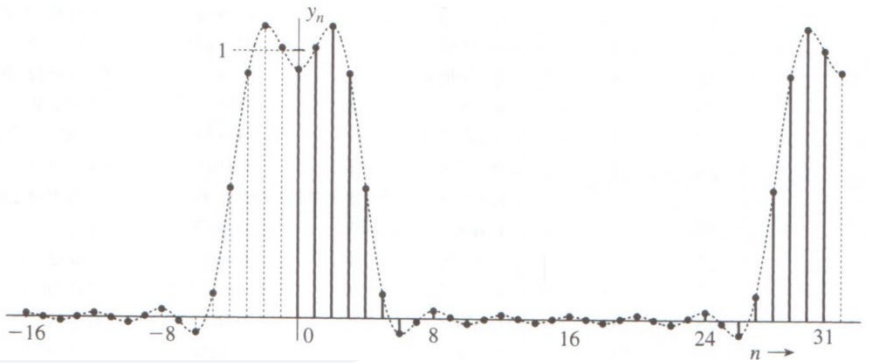
- Let x_n be sampled signal with N_1 samples
- Let g_n be sampled signal with N_2 samples
- The convolution $c_n = x_n * g_n$ will have N_1+N_2-1 values
- To use DFT to find convolution, need to make sure have right number of values – pad x_n and g_n with zeros
- Process:
  1. Find DFTs X_r and G_r of suitably padded x_n and g_n
  2. Multiply X_r by G_r
  3. Find IDFT of $X_r G_r$

Applications of DFT – Filtering

- Get DFT of input signal (X_r)
- Multiply X_r by H_r (samples of filter $H(j\omega)$)
  - Spectrum of X_r is shaped (filtered) by H_r
- Take IDFT of X_r H_r to get filtered output signal y_n







Properties of DFT

- Linear

If $x_n \Leftrightarrow X_r$ and $g_n \Leftrightarrow G_r$, then

$$a_1 x_n + a_2 g_n \Leftrightarrow a_1 X_r + a_2 G_r$$

- Conjugate Symmetry

When $x(t)$ real,

$$X_r^* = X_{-r}$$

and since X_r is N_o periodic,

$$X_r^* = X_{N_o - r}$$

- Therefore only need to compute half the values, rest are conjugate

Properties of DFT

- Time Shifting

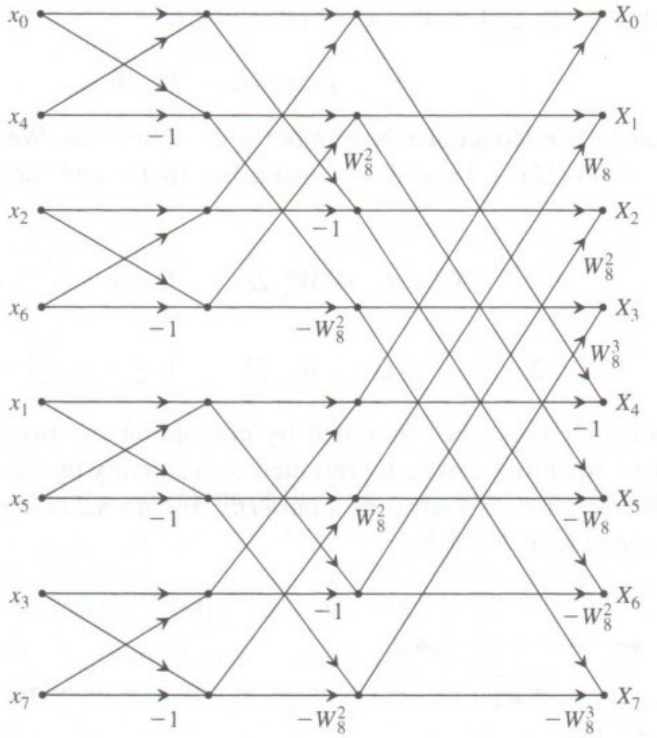
$$x_{n-k} \Leftrightarrow X_r \, e^{-jr\Omega_0 k}$$

- Frequency Shifting

$$x_n e^{jr\Omega_0 m} \Leftrightarrow X_{r-m}$$

Fast Fourier Transform (FFT)

- DFT calculated using linearity and symmetry properties to save on calculations
- No loss of accuracy compared to DFT



Steps in 8-point FFT

Take Away

- After studying this material, you should be able to:

1. Understand what spectral sampling is
2. Understand what a Discrete Fourier Transform is and how it can be used
3. Understand aliasing, leakage and spectral spreading in DFTs
4. Choose appropriate parameters for DFT
5. Work out basic discrete waveforms and spectra associated with DFT
6. Understand what a Fast Fourier Transform is